Year 2 Pure Mathematics

Chapter 6

Trigonometric functions

Practice Booklet

Contents

6.1 Secant, cosecant and cotangent ..... book p. 43
6.2 Graphs of sec x, cosec x and cot x ..... book p. 43
6.3 Using sec x, cosec x and cot x ..... book p. 45
6.4 Trigonometric identities ..... book p. 45
6.5 Inverse trigonometric functions ..... book p. 46
Problem solving: Set A ..... book p. 47
Problem solving: Set B ..... book p. 48
Selected exam question bank questions .....
Answers .....

6.1 Secant, cosecant and cotangent

Practice Book, printed page 43.

1. Without using your calculator, work out the exact values of:

a. $\sec 30°$

b. $\operatorname{cosec} \frac{5\pi}{4}$

c. $\cot \frac{7\pi}{6}$

Hint

$\sec x = \frac{1}{\cos x}$, $\operatorname{cosec} x = \frac{1}{\sin x}$, $\cot x = \frac{1}{\tan x}$ and $\cot x = \frac{\cos x}{\sin x}$

2. Use your calculator to find the values of the following, giving your answers to 3 significant figures:

a. $\cot 3.6$ rad

b. $\operatorname{cosec} 153°$

c. $\sec \frac{7\pi}{4}$

Hint

Your calculator may not have keys for these functions. For example, for part a, find $\tan 3.6$ rad, then find its reciprocal.

3. Find the exact values of:

a. $\sec 150°$

b. $\cot(-135°)$

c. $\operatorname{cosec} \frac{3\pi}{2}$

4. P Show that:

a. $\frac{\sec 60°}{\operatorname{cosec} 270°} = -2$

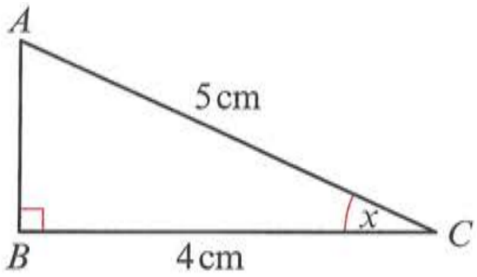
b. $\operatorname{cosec} \frac{3\pi}{4} \cot \frac{\pi}{6} = \sqrt{6}$

5. P Prove that:

a. $\sec(-\theta) \equiv \sec \theta$

b. $\cot(\theta + \pi) \equiv \cot \theta$

6. P ABC is a right-angled triangle, with angle $ACB = x$, and sides $AC = 5$ cm and $BC = 4$ cm.



Write down the exact value of:

a. $\sec x$

b. $\operatorname{cosec} x$

c. $\cot x$

7. E/P Show that $\cot \frac{\pi}{3} - \operatorname{cosec} \frac{\pi}{4} = \frac{\sqrt{a} - a\sqrt{b}}{a}$, where a and b are integers to be found. (2 marks)

6.2 Graphs of $\sec x$, $\operatorname{cosec} x$ and $\cot x$

Practice Book, printed page 43.

1.

a.

i. On the same set of axes, sketch the graphs of $y = \sec x$ and $y = 3$ in the interval $0 \leq x \leq 2\pi$.

ii. State the number of solutions to the equation $\sec x - 3 = 0$ for $0 \leq x \leq 2\pi$.

Hint

The number of solutions to the equation is the same as the number of points of intersection of the graphs.
The graph of $y = \sec x$ is undefined for values of x for which $\cos x = 0$. It has vertical asymptotes at these points.

b.

i. On the same set of axes, sketch the graphs of $y = \operatorname{cosec} x$ and $y = -\sin x$ in the interval $-180° \leq x \leq 180°$

ii. Explain how your graphs show that there are no real solutions to $\operatorname{cosec} x = -\sin x$.

Hint

The graph of $y = \operatorname{cosec} x$ is undefined for values of x for which $\sin x = 0$, and has vertical asymptotes at these points.

2.

a.

i. On separate sets of axes, sketch the graphs of $y = \cot x$ and $y = -\tan x$ in the interval $-360° \leq x \leq 360°$.

ii. Hence state a relationship between $\cot x$ and $-\tan x$.

Hint

The graph of $y = \cot x$, is undefined for values of x for which $\tan x = 0$, and has vertical asymptotes at these points.

b.

i. On the same set of axes, sketch the graphs of $y = \sec x$ and $y = \operatorname{cosec} x$ in the interval $-\pi \leq x \leq \frac{3\pi}{2}$

ii. Hence state a relationship between $\sec x$ and $\operatorname{cosec} x$.

3. Sketch the following graphs on separate sets of axes in the interval $0 \leq x \leq 360°$, showing the

coordinates of any maximum and minimum points, and any points of intersection with the coordinate axes.

a. $y = 1 + \operatorname{cosec} x$

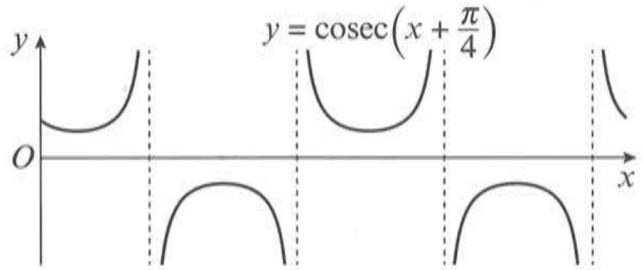
b. $y = -\cot \frac{1}{2}x$

c. $y = 3 \sec 2x$

Hint

Start by sketching the graph of $\sec x$, $\operatorname{cosec} x$ or $\cot x$, and then apply each stretch or translation to transform the graph.

4. P The graph shows the curve $y = \operatorname{cosec}\left(x + \frac{\pi}{4}\right)$ in the interval $0 \leq x \leq 4\pi$.



Write down:

a. the coordinates of the y-intercept

b. the turning points of the curve

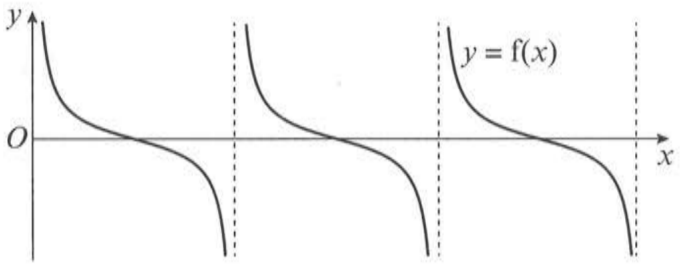
c. the equations of any vertical asymptotes.

5. E/P

a. Sketch the graph of $y = 2 + 4 \sec x$ in the interval $-360° \leq x \leq 360°$. (3 marks)

b. Given that the equation $2 + 4 \sec x = a$ has no real solutions, find the range of possible values for a. (1 mark)

6. E/P The diagram shows a sketch of the curve $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = \cot 3x$, for $0 \leq x < 180°$, $x \neq 60°, 120°$.



a. Write down the number of solutions to the equation $\mathrm{f}(x) = x$. (1 mark)

b. Write down the solutions to the equation $\mathrm{f}(x) = 0$. (2 marks)

7. E/P

a. On the same set of axes, sketch the graphs of $y = 2 + 3 \operatorname{cosec} x$ and $y = -x$ in the interval $-\pi \leq x \leq \pi$. (3 marks)

b. Write down the number of solutions to the equation $2 + 3 \operatorname{cosec} x + x = 0$ in the interval $-\pi \leq x \leq \pi$. (1 mark)

c. Deduce the maximum value of $\dfrac{1}{2 + 3 \operatorname{cosec} x}$ and give the smallest positive value of x at which this occurs. (2 marks)

6.3 Using $\sec x$, $\operatorname{cosec} x$ and $\cot x$

Practice Book, printed page 45.

1. Write down the value of:

a. $\sec x$ given that $\cos x = \dfrac{1}{4}$

b. $\operatorname{cosec} x$ given that $\dfrac{3}{\sin x} = 5$

c. $\cot x$ given that $2\cos x = 3\sin x$

Hint

For part c, use $\dfrac{\sin x}{\cos x} = \tan x$

← Year 1, Section 10.3

2. Simplify each expression, writing your answer as a single trigonometric function.

a. $\cos x \cot x + \sin x$

b. $\dfrac{1}{\cos x \tan x}$

c. $\cot 3x \sec 3x$

Hint

Start by writing $\sec x$, $\operatorname{cosec} x$ or $\cot x$ in terms of $\sin x$, $\cos x$ or $\tan x$ and then simplify.

3. Solve the following equations for values of x in the interval $0 \leq x \leq 360°$, giving your answers to 3 significant figures where necessary.

a. $\operatorname{cosec} x = 2$

b. $\sqrt{3}\sec x = 4$

c. $3\cot x = -2.5$

Hint

Write the equivalent equation for $\sin x$, $\cos x$ or $\tan x$, then solve.

4. E/P

a. Prove that $\dfrac{\cot x + \tan x}{\sec x} \equiv \operatorname{cosec} x$ (4 marks)

b. Hence solve in the interval $0 \leq x \leq 360°$, the equation $\dfrac{\cot x + \tan x}{\sec x} = 4$ (3 marks)

5. E/P

a. Prove that $\cos 3\theta + \sin 3\theta \tan 3\theta \equiv \sec 3\theta$ (4 marks)

b. Explain why the equation $2\cos 3\theta + 2\sin 3\theta \tan 3\theta = -1$ has no solutions. (2 marks)

6. E/P Solve $3\sin x = 8\cot x$ in the interval $0 \leq x \leq 2\pi$. (4 marks)

7. E/P

a. Prove that $\dfrac{\sin x}{1+\cos x} + \dfrac{1+\cos x}{\sin x} \equiv 2\operatorname{cosec} x$ (4 marks)

b. Hence solve in the interval $0 \leq x \leq 2\pi$, the equation

$$\frac{\sin\left(x - \frac{\pi}{6}\right)}{1+\cos\left(x - \frac{\pi}{6}\right)} + \frac{1+\cos\left(x - \frac{\pi}{6}\right)}{\sin\left(x - \frac{\pi}{6}\right)} = 5$$

(4 marks)

6.4 Trigonometric identities

Practice Book, printed page 45.

1. Show that:

a. $\operatorname{cosec}^2 x + \tan^2 x \equiv \sec^2 x + \cot^2 x$

b. $\cot^2 x + \cos^2 x \equiv (\operatorname{cosec} x - \sin x)(\operatorname{cosec} x + \sin x)$

c. $\dfrac{1}{1+\sin x} + \dfrac{1}{1-\sin x} \equiv 2 + 2\tan^2 x$

Hint

Use these identities:
$1 + \tan^2 x \equiv \sec^2 x$
$1 + \cot^2 x \equiv \operatorname{cosec}^2 x$
$\sin^2 x + \cos^2 x \equiv 1$

2. Given that $\sin x = \dfrac{4}{5}$ and that $90° \leq x \leq 180°$, find the exact values of:

a. $\cos x$

b. $\cot x$

c. $\operatorname{cosec} x$

Hint

The interval $90° \leq x \leq 180°$ means that x lies in the second quadrant, where $\cos x$ and $\tan x$ are negative.

3. Solve these equations in the given intervals.
Give your answers to 3 significant figures when they are not exact.

a. $\sec^2 x + \tan x = 1$ for $-\pi \leq x \leq \pi$

b. $3\cot^2 x + 9\operatorname{cosec} x + 1 = 0$ for $0 \leq x \leq 2\pi$

c. $\operatorname{cosec}^2 x + \cot^2 x = 3$ for $0 \leq x \leq 2\pi$

Hint

Use the identities $1 + \tan^2 x = \sec^2 x$ and $1 + \cot^2 x = \operatorname{cosec}^2 x$ to write each equation in terms of one trigonometrical function only, and then factorise and solve.

4. E/P Given that $7\cot^2 x - 5\operatorname{cosec}^2 x = 2$, and that x is obtuse, find the exact value of $\cos x$. (3 marks)

5. E/P Solve $\sec^2 x = 5 + 3\tan x$ in the interval $0 \leq x \leq 360°$, giving your answers to 1 decimal place where necessary. (4 marks)

6. E/P Given that $\cot^2 p = -4\operatorname{cosec} p$,

a. find the value of $\operatorname{cosec} p$ (4 marks)

b. show that $\sin p = 2 - \sqrt{5}$. (2 marks)

c. Hence solve $\cot^2 p = -4\operatorname{cosec} p$ in the interval $0 \leq p \leq 2\pi$, giving your answers to 1 decimal place. (3 marks)

7. E/P Solve $2\operatorname{cosec}^2 x + 5\cot x = 5$ in the interval $-\pi \leq x \leq \pi$, giving your answers to 2 decimal places. (4 marks)

6.5 Inverse trigonometric functions

Practice Book, printed page 46.

1. In each of the following cases, sketch $y = \mathrm{f}(x)$, showing any points where the curve intersects the coordinate axes, and the domain and range of each function.

a. $\mathrm{f}(x) = \arcsin x$

b. $\mathrm{f}(x) = \arccos x$

c. $f(x) = \arctan x$

Hint

$\arcsin x$, $\arccos x$ and $\arctan x$ are the inverse functions of $\sin x$, $\cos x$ and $\tan x$ respectively. These inverse functions have a restricted domain to ensure they are one-to-one functions.

2. Without using a calculator work out the following, giving your answer in terms of π.

a. $\arctan(1)$

b. $\arcsin\left(-\frac{\sqrt{3}}{2}\right)$

c. $\arccos\left(\frac{1}{2}\right)$

Hint

You can work these out without a calculator by considering exact values of sin, cos and tan.

3. Without using a calculator work out the values of:

a. $\arcsin\left(\sin\frac{\pi}{2}\right)$

b. $\arccos(\cos 3\pi)$

c. $\arctan\left(\tan\frac{5\pi}{4}\right)$

Hint

Make sure you give your answers in the appropriate ranges for arcsin, arccos and arctan.

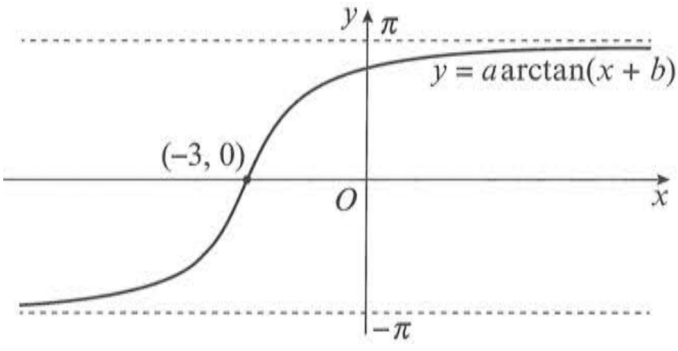
4. E/P

a. Sketch the graph of $y = \frac{\pi}{3} + \arcsin\left(x - \frac{1}{2}\right)$, clearly showing the exact coordinates of the endpoints of the curve. (4 marks)

b. Find the exact coordinates of the point where the curve crosses the x-axis. (3 marks)

5. E/P Find in terms of π, the value of $\arccos\left(\frac{\sqrt{3}}{2}\right) + \arccos\left(-\frac{1}{2}\right)$ (2 marks)

6. E/P The diagram shows the graph of $y = a\arctan(x + b)$, where a and b are constants to be determined.



a. Find a and b. (3 marks)

b. Find the coordinates of the point where the curve crosses the y-axis, giving your answer to 3 decimal places. (2 marks)

7. E/P Given that $\arccos(x - 3) = -\frac{\pi}{4}$, find the exact value of x. (3 marks)

Problem solving: Set A

Bronze

a. Given that $\dfrac{\cos x}{1-\sin x}+\dfrac{1-\sin x}{\cos x}=p$, show that
$\dfrac{\cos^2 x+(1-2\sin x+\sin^2 x)}{(1-\sin x)(\cos x)}=p$ (2 marks)

b. Hence show that $\sec x=\dfrac{p}{2}$ (3 marks)

Silver

a. Given that $\cot^2 x+\operatorname{cosec}^2 x=a$, show that $\operatorname{cosec}^2 x=\dfrac{a+1}{2}$ (2 marks)

b. Hence show that $\sec^2 x=\dfrac{a+1}{a-1}$, $a\neq 1$. (3 marks)

Gold

Given that $\dfrac{1-\tan^2 x}{1+\tan^2 x}=z$, show that $\operatorname{cosec}^2 x=\dfrac{2}{1-z}$, $z\neq 1$. (5 marks)

Problem solving: Set B

Bronze

a.

i. Sketch the curve $y=\tan 2x$ in the interval $0\leq x\leq \pi$, showing the points where the curve crosses the x-axis.

ii. Write down the period of $\tan 2x$.

(3 marks)

b.

i. Show that the equation $\cot^2 2x-4\cot 2x+4=0$ can be written as $(\cot 2x-a)^2=0$, where a is a constant to be found.

ii. Write down the value of $\tan 2x$, and hence solve the equation $\cot^2 2x-4\cot 2x=-4$ in the interval $0\leq x\leq \pi$, giving your answers to 2 decimal places.

(4 marks)

Silver

a. Given $4\cot^2 2x+12\operatorname{cosec} 2x+9=0$, show that $\sin 2x=-\dfrac{2}{5}$ (4 marks)

b. Hence solve $4\cot^2 2x+12\operatorname{cosec} 2x+9=0$ in the interval $0\leq x\leq 360°$, giving your answers to 1 decimal place. (3 marks)

Gold

Solve $5+2\cot^2 a=7(1+\operatorname{cosec} a)$, where $a=2x+\dfrac{\pi}{3}$, in the interval $0\leq x\leq 2\pi$.
Give your answers to 2 decimal places. (7 marks)

Now try this ⟶ Exam question bank Q3, Q6, Q15, Q21, Q42, Q58

Selected exam question bank questions

Question 3

Given that $m = 2\operatorname{cosec} x$ and $n = \frac{1}{5}\sin x$, express m in terms of n. (4 marks)

Question 6

Solve $4\cos x = 3\sec x$ in the interval $-180° \leq x \leq 180°$. (4 marks)

Question 15

a. On the same set of axes, sketch the graphs of $y = 3 + 3\sec x$ and $y = \cot x$ in the interval $-180° \leq x \leq 180°$. (3 marks)

b. Write down the number of solutions to the equation $3 + 3\sec x - \cot x = 0$ in the interval $-180° \leq x \leq 180°$. (1 mark)

c. Write down the maximum value of $\dfrac{1}{3 + 3\sec x}$ (1 mark)

Question 21

a. Given that $\cot P = -2$, where $\pi \leq P \leq 2\pi$, calculate the exact value of $\sin P$. (3 marks)

b. Show that $\sec P = \pm\dfrac{\sqrt{a}}{2}$, where a is an integer to be found. (3 marks)

Question 42

a. Prove that $\sec^4 x - \tan^4 x \equiv 2\sec^2 x - 1$. (4 marks)

b. Hence solve the equation $\sec^4 x - \tan^4 x = 24$ in the interval $0 \leq x \leq 360°$.
Give your answers to 4 significant figures. (4 marks)

Question 58

The function g is defined as $\mathrm{g}(x) = \arccos x$, for $-1 \leq x \leq 1$.

The function h is defined as $\mathrm{h}(x) = \frac{1}{2}x$, $x \in \mathbb{R}$.

a. Find an expression for $\mathrm{gh}(x)$ and state its domain. (3 marks)

b. Sketch the graph of $y = \mathrm{gh}(x)$, clearly showing the exact coordinates of the endpoints of the curve. (3 marks)

c. Determine $(\mathrm{gh})^{-1}(x)$ and state its domain. (2 marks)

Answers

6.1 Secant, cosecant and cotangent

1.

a. $\frac{2\sqrt{3}}{3}$

b. $-\sqrt{2}$

c. $\sqrt{3}$

2.

a. 2.03

b. 2.20

c. 1.41

3.

a. $-\frac{2\sqrt{3}}{3}$

b. 1

c. -1

4.

a. $\frac{\sec 60^\circ}{\operatorname{cosec} 270^\circ} = \frac{2}{-1} = -2$

b. $\operatorname{cosec}\left(\frac{3\pi}{4}\right)\cot\left(\frac{\pi}{6}\right) = \sqrt{2} \times \sqrt{3} = \sqrt{6}$

5.

a. $\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\cos(-\theta)} = \sec(-\theta)$

b. $\cot(\theta + \pi) = \frac{1}{\tan(\theta + \pi)} = \frac{1}{\tan\theta} = \cot\theta$

6.

a. $\frac{5}{4}$

b. $\frac{5}{3}$

c. $\frac{4}{3}$

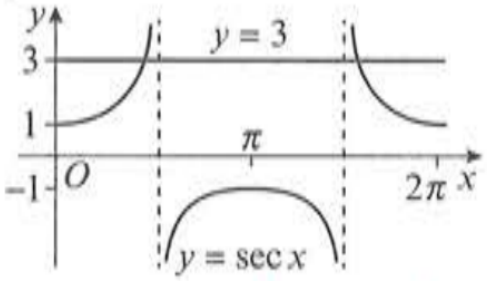
7. $\cot\left(\frac{\pi}{3}\right) - \operatorname{cosec}\left(\frac{\pi}{4}\right) = \frac{\sqrt{3}}{3} - \sqrt{2} = \frac{\sqrt{3} - 3\sqrt{2}}{3}$

6.2 Graphs of $\sec x$, $\operatorname{cosec} x$ and $\cot x$

1.

a.

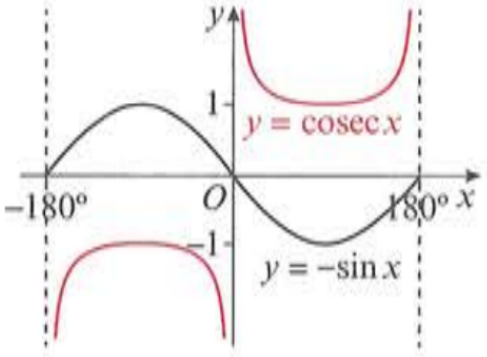
i.



ii. 2

b.

i.

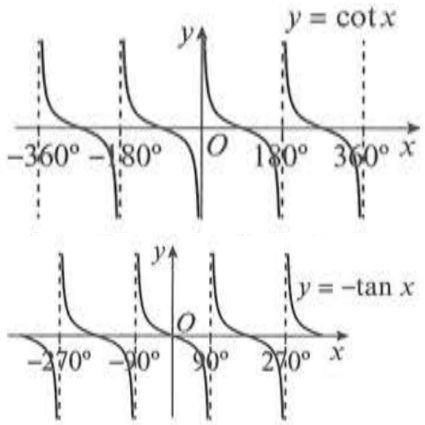


ii. Graphs do not intersect.

2.

a.

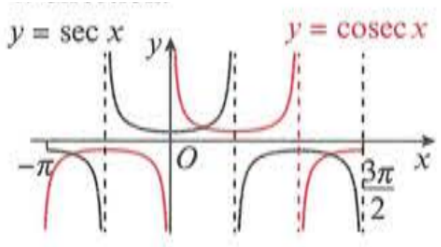
i.



ii. The graph of $y = \cot x$ is the same as that of $y = -\tan x$ translated by 90° in the positive or negative x-direction.

b.

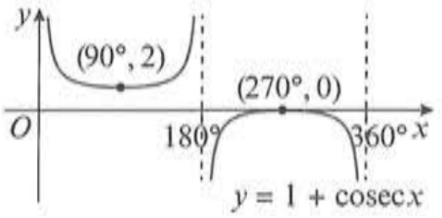
i.



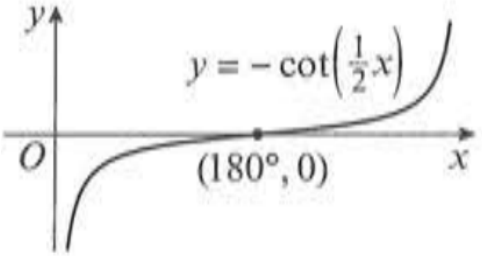
ii. The graph of $\operatorname{cosec} x$ is the same as that of $\sec x$ translated by $\frac{\pi}{2}$ in the positive x-direction.

3.

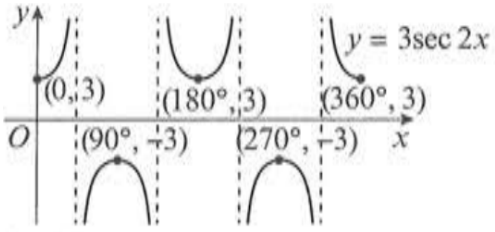
a.



b.



c.



4.

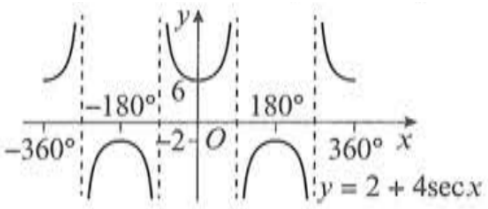
a. $(0, \sqrt{2})$

b. $\left(\frac{\pi}{4}, 1\right), \left(\frac{9\pi}{4}, 1\right), \left(\frac{5\pi}{4}, -1\right), \left(\frac{13\pi}{4}, -1\right)$

c. $x = \frac{3\pi}{4}, x = \frac{7\pi}{4}, x = \frac{11\pi}{4}, x = \frac{15\pi}{4}$

5.

a.



b. $-2 < a < 6$

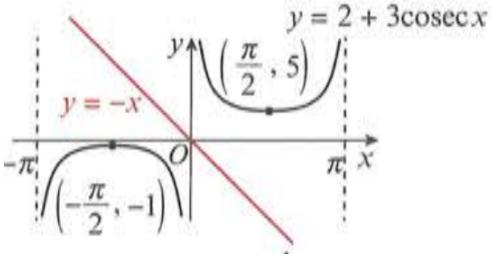
6.

a. 3

b. $x = 30°, 90°, 150°$

7.

a.



b. 0

c. $\frac{1}{5}$ when $x = \frac{\pi}{2}$

6.3 Using $\sec x$, $\operatorname{cosec} x$ and $\cot x$

1.

a. 4

b. $\frac{5}{3}$

c. $\frac{3}{2}$

2.

a. $\operatorname{cosec} x$

b. $\operatorname{cosec} x$

c. $\operatorname{cosec} 3x$

3.

a. $x = 30°, 150°$

b. $x = 64.3°, 296°$

c. $x = 130°, 310°$

4.

a. $\dfrac{\cot x + \tan x}{\sec x} \equiv \dfrac{\frac{\cos x}{\sin x} + \frac{\sin x}{\cos x}}{\frac{1}{\cos x}}$

$\equiv \dfrac{\cos^2 x}{\sin x} + \sin x \equiv \dfrac{1 - \sin^2 x}{\sin x} + \sin x$

$\equiv \dfrac{1 - \sin^2 x + \sin^2 x}{\sin x} \equiv \operatorname{cosec} x$

b. $x = 14.5°, 166°$ (3 s.f.)

5.

a. $\cos 3\theta + \sin 3\theta \times \dfrac{\sin 3\theta}{\cos 3\theta} = \cos 3\theta + \dfrac{1 - \cos^2 3\theta}{\cos 3\theta}$

$= \dfrac{\cos^2 3\theta + 1 - \cos^2 3\theta}{\cos 3\theta} \equiv \sec 3\theta$

b. $2\cos 3\theta + 2\sin 3\theta \tan 3\theta = -1$
$\Rightarrow 2(\cos 3\theta + \sin 3\theta \tan 3\theta) = -1 \Rightarrow 2\sec 3\theta = -1$
$\Rightarrow \sec 3\theta = -\dfrac{1}{2}$
$\sec 3\theta = k$ has no solutions in the range $-1 < k < 1$

6. $x = 1.23, 5.05$ (3 s.f.)

7.

a. L.H.S. $\equiv \dfrac{(\sin x)(\sin x) + (1+\cos x)(1+\cos x)}{(1+\cos x)\sin x}$
$\equiv \dfrac{(1-\cos^2 x) + 1 + 2\cos x + \cos^2 x}{(1+\cos x)\sin x}$
$\equiv \dfrac{2(1+\cos x)}{(1+\cos x)\sin x} \equiv 2\operatorname{cosec} x$

b. $x = 0.935, 3.25$ (3 s.f.)

6.4 Trigonometric identities

1.

a. $\operatorname{cosec}^2 x + \tan^2 x \equiv (1 + \cot^2 x) + (\sec^2 x - 1)$
$\equiv \sec^2 x + \cot^2 x$

b. $\cot^2 x + \cos^2 x \equiv (\operatorname{cosec}^2 x - 1) + (1 - \sin^2 x)$
$\equiv (\operatorname{cosec} x - \sin x)(\operatorname{cosec} x + \sin x)$

c. $\dfrac{1}{1+\sin x} + \dfrac{1}{1-\sin x} \equiv \dfrac{1 - \sin x + 1 + \sin x}{(1+\sin x)(1-\sin x)}$
$\equiv \dfrac{2}{1-\sin^2 x} \equiv 2\sec^2 x \equiv 2 + 2\tan^2 x$

2.

a. $-\dfrac{3}{5}$

b. $-\dfrac{3}{4}$

c. $\dfrac{5}{4}$

3.

a. $x = -\pi, -\dfrac{\pi}{4}, 0, \dfrac{3\pi}{4}, \pi$

b. $x = 3.46, 5.97$

c. $x = \dfrac{\pi}{4}, \dfrac{3\pi}{4}, \dfrac{5\pi}{4}, \dfrac{7\pi}{4}$

4. $\cos x = -\dfrac{\sqrt{7}}{3}$

5. $x = 76.0°, 135°, 256°, 315°$

6.

a. $-2 - \sqrt{5}$

b. $\sin p = \dfrac{1}{-2-\sqrt{5}} = \dfrac{-2+\sqrt{5}}{(-2-\sqrt{5})(-2+\sqrt{5})} = 2 - \sqrt{5}$

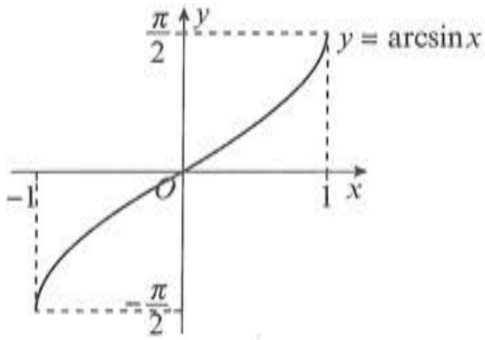
c. $p = 3.4, 6.0$

7. $x = -2.03, -0.32, 1.11, 2.82$

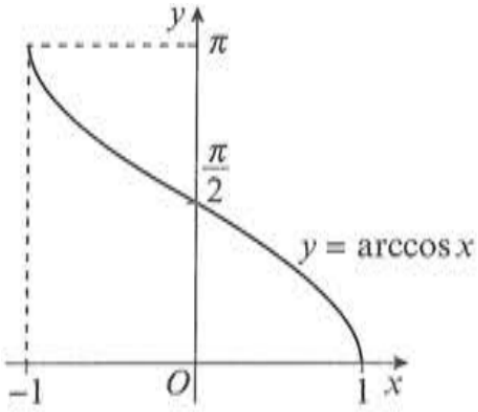
6.5 Inverse trigonometric functions

1.

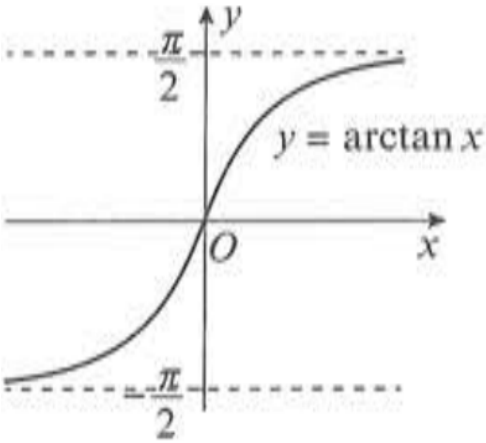
a.



b.



c.



2.

a. $\frac{\pi}{4}$

b. $-\frac{\pi}{3}$

c. $\frac{\pi}{3}$

3.

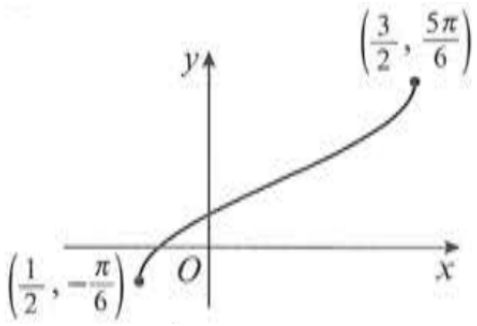
a. $\frac{\pi}{2}$

b. π

c. $\frac{\pi}{4}$

4.

a.



b. $\left(\frac{1-\sqrt{3}}{2}, 0\right)$

5. $\frac{5\pi}{6}$

6.

a. $a = 2, b = 3$

b. $(0, 2.498)$

7. $\frac{6+\sqrt{2}}{2}$

Problem solving: Set A

Bronze

a. $p = \frac{\cos x}{1-\sin x} + \frac{1-\sin x}{\cos x} = \frac{\cos^2 x + (1-\sin x)^2}{(1-\sin x)\cos x}$
$= \frac{\cos^2 x + (1 - 2\sin x + \sin^2 x)}{(1-\sin x)\cos x}$

b. $p = \frac{2-2\sin x}{(1-\sin x)\cos x}$
$= \frac{2(1-\sin x)}{(1-\sin x)\cos x} = \frac{2}{\cos x} = 2\sec x$
$\Rightarrow 2\sec x = p \Rightarrow \sec x = \frac{p}{2}$

Silver

a. $\cot^2 x + \operatorname{cosec}^2 x = a \Rightarrow (\operatorname{cosec}^2 x - 1) + \operatorname{cosec}^2 x = a$
$\Rightarrow 2\operatorname{cosec}^2 x - 1 = a \Rightarrow \operatorname{cosec}^2 x = \frac{a+1}{2}$

b. $\operatorname{cosec}^2 x = \frac{a+1}{2} \Rightarrow \sin^2 x = \frac{2}{a+1}$
$\Rightarrow 1 - \cos^2 x = \frac{2}{a+1} \Rightarrow \cos^2 x = 1 - \frac{2}{a+1}$
$\Rightarrow \cos^2 x = \frac{a-1}{a+1} \Rightarrow \sec^2 x = \frac{a+1}{a-1}, a \neq 1$

Gold

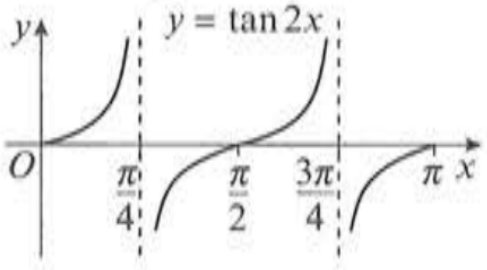
$z = \frac{1-\tan^2 x}{1+\tan^2 x} = \frac{1-\tan^2 x}{\sec^2 x} = \cos^2 x \left(1 - \frac{\sin^2 x}{\cos^2 x}\right)$
$= \cos^2 x - \sin^2 x = (1 - \sin^2 x) - \sin^2 x$
$= 1 - 2\sin^2 x$
$\Rightarrow \sin^2 x = \frac{1-z}{2} \Rightarrow \operatorname{cosec}^2 x = \frac{2}{1-z}$

Problem solving: Set B

Bronze

a.

i.



ii. $\frac{\pi}{2}$

b.

i. Factorise $\cot^2 2x - 4\cot 2x + 4 = 0$
to give $(\cot 2x - 2)^2 = 0$, $a = 2$

ii. $\tan 2x = \frac{1}{2}$, solutions: $x = 0.23, 1.80$

Silver

a. $4(\operatorname{cosec}^2 2x - 1) + 12\operatorname{cosec} 2x + 9 = 0$
$4\operatorname{cosec}^2 2x + 12\operatorname{cosec} 2x + 5 = 0$
$(2\operatorname{cosec} 2x + 1)(2\operatorname{cosec} 2x + 5) = 0$
$\operatorname{cosec} 2x = -\frac{1}{2}$ (no solutions) and $\operatorname{cosec} 2x = -\frac{5}{2}$
Therefore $\sin 2x = -\frac{2}{5}$

b. $x = 101.8°, 168.2°, 281.8°, 348.2°$

Gold $x = 0.92, 2.74, 4.06, 5.89$

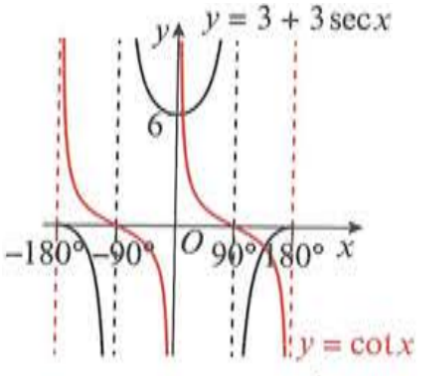
Selected exam question bank answers

Question 3 $m = \frac{2}{5n}$

Question 6 $x = -150°, -30°, 30°, 150°$

Question 15

a.



b. 2

c. $\frac{1}{6}$

Question 21

a. $\pm\dfrac{1}{\sqrt{5}}$

b. $\cos P = \pm\dfrac{2}{\sqrt{5}} \Rightarrow \sec P = \pm\dfrac{\sqrt{5}}{2}$

Question 42

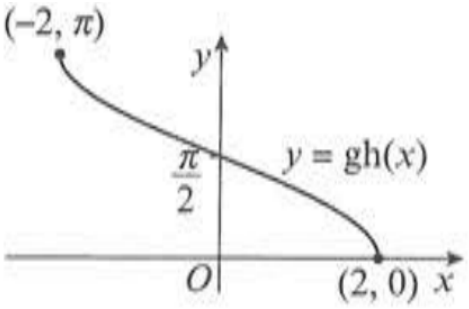
a. $(\sec^2 x + \tan^2 x)(\sec^2 x - \tan^2 x)$
$\equiv (\sec^2 x + \sec^2 x - 1)(\sec^2 x - \sec^2 x + 1)$
$\equiv (2\sec^2 x - 1)(1) \equiv 2\sec^2 x - 1$

b. $x = 73.57°, 106.4°, 253.6°, 286.4°$

Question 58

a. $\text{gh}(x) = \arccos\left(\dfrac{1}{2}x\right),\ -2 \leq x \leq 2$

b.



c. $(\text{gh})^{-1}(x) = 2\cos x,\ 0 \leq x \leq \pi$